

Machine Learning Mini Project

Project Title

AI Startup Success Predictor using Machine Learning

1. Project Overview

The startup ecosystem is growing rapidly across the world. Every year, thousands of startups are launched, but only a small percentage survive for more than five years. Investors, venture capitalists, banks, incubators, and entrepreneurs often need to evaluate whether a startup has the potential to succeed before investing time and money.

The objective of this project is to develop a **Machine Learning model** that predicts the probability of a startup surviving for the next five years based on various business-related factors.

The project will involve collecting data, preprocessing it, performing exploratory data analysis, training different machine learning models, evaluating their performance, and deploying the best-performing model as a web application.

2. Problem Statement

Develop a Machine Learning model that predicts whether a startup is likely to survive for the next five years based on its funding, founders, market conditions, competition, team size, and other business parameters.

3. Objectives

After completing this project, students should be able to:

- Understand the complete Machine Learning lifecycle.
- Collect and preprocess real-world datasets.
- Perform Exploratory Data Analysis (EDA).
- Apply feature engineering techniques.
- Train and compare multiple Machine Learning algorithms.
- Evaluate model performance using different metrics.
- Deploy the trained model using Streamlit or Flask.
- Present business insights obtained from the model.

4. Expected Learning Outcomes

Students will gain hands-on experience in:

- Data Collection
- Data Cleaning
- Feature Engineering
- Exploratory Data Analysis
- Data Visualization
- Classification Algorithms
- Hyperparameter Tuning
- Model Evaluation
- Model Deployment
- Report Writing

5. Dataset Sources

Students may use any one or a combination of the following sources:

Option 1: Kaggle

Search for datasets such as:

- Startup Success Prediction
- Startup Investments
- Startup Funding Dataset
- Venture Capital Dataset
- Unicorn Companies Dataset

Option 2: Crunchbase

Collect publicly available startup information such as:

- Funding
- Founders
- Industry
- Investors
- Headquarters
- Company Status

Option 3: Startup India

Collect information about Indian startups including:

- Startup Name
- Sector
- Registration Year
- State
- Industry

Option 4: Create Your Own Dataset

Students may combine information from multiple public sources to prepare their own dataset.

Minimum Dataset Size:

- 1,000 records (recommended)
- At least 15 features

6. Input Features

The dataset may contain the following attributes.

Feature	Description
Startup Name	Company Name
Funding Amount	Total Funding Raised
Number of Founders	Total Founders
Founder Experience	Average Years of Experience
Industry	Startup Sector
Team Size	Number of Employees
Market Size	Small/Medium/Large
Competition Level	Low/Medium/High
Revenue Growth	Annual Growth Rate
Burn Rate	Monthly Expenses
Customer Rating	Average Rating
Number of Customers	Active Customers
Customer Retention	Percentage
Patent Count	Total Patents
Investor Type	Angel/VC/Bootstrapped

Target Variable

Startup Status

- 1 = Successful
- 0 = Failed

7. Software Requirements

Programming Language

- Python 3.11 or above

IDE

- Jupyter Notebook
- Google Colab
- VS Code
- PyCharm

Python Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- XGBoost
- Plotly
- Streamlit

8. Project Workflow

Data Collection



Data Cleaning



Exploratory Data Analysis



Feature Engineering



Data Preprocessing



Model Training



Model Evaluation



Model Comparison



Deployment



Final Report

9. Step-by-Step Implementation

Step 1: Data Collection

Collect startup information from one or more data sources.

Deliverables:

- CSV Dataset
- Dataset Description

Step 2: Data Understanding

Study the dataset.

Check:

- Number of rows
- Number of columns
- Data types
- Missing values
- Duplicate records

Functions to use

- `head()`
- `info()`
- `describe()`
- `shape`
- `isnull()`
- `duplicated()`

Step 3: Data Cleaning

Perform the following:

- Remove duplicate records
- Handle missing values
- Remove unnecessary columns

- Correct inconsistent values
- Handle outliers

Document every preprocessing step.

Step 4: Exploratory Data Analysis (EDA)

Perform graphical analysis.

Suggested graphs:

- Histogram
- Box Plot
- Pie Chart
- Bar Graph
- Scatter Plot
- Correlation Heatmap
- Pair Plot

Analyze:

- Funding distribution
- Industry distribution
- Success rate
- Team size
- Founder experience
- Competition level

Step 5: Feature Engineering

Create new useful features.

Examples:

Funding per Employee

Founder Experience Score

Revenue Growth Rate

Burn Ratio

Customer Growth Percentage

Step 6: Data Encoding

Convert categorical variables.

Examples:

Industry

Competition

Investor Type

Use:

- Label Encoding
- One-Hot Encoding

Step 7: Feature Scaling

Normalize numerical values.

Use:

- StandardScaler
- MinMaxScaler

Step 8: Split Dataset

Split the dataset into:

- Training Data (80%)
- Testing Data (20%)

Use:

`train_test_split()`

10. Machine Learning Models

Students must implement at least **four** algorithms.

Model 1

Logistic Regression

Purpose:

Baseline classification model.

Model 2

Decision Tree

Purpose:

Understand decision-making process.

Model 3

Random Forest

Purpose:

Improve prediction accuracy using multiple decision trees.

Model 4

XGBoost

Purpose:

Improve prediction by correcting mistakes made by previous trees.

Advantages:

- High accuracy
- Fast execution
- Handles missing values
- Prevents overfitting
- Excellent for structured datasets

11. Model Evaluation

Evaluate every model using:

- Accuracy

- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

Compare all models in a table.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression				
Decision Tree				
Random Forest				
XGBoost				

Select the best-performing model with proper justification.

12. Feature Importance Analysis

Using Random Forest or XGBoost:

Identify:

- Most important features
- Least important features

Visualize feature importance using a bar chart.

Discuss how these features influence startup success.

13. Model Deployment

Develop a simple web application using **Streamlit**.

Input fields:

- Funding Amount
- Founder Experience
- Team Size
- Industry
- Competition Level
- Market Size

Output:

- Probability of Success
- Prediction (Successful/Not Successful)
- Confidence Score

14. Project Deliverables

Each student/group must submit:

1. Dataset (CSV)
2. Jupyter Notebook (.ipynb)
3. Python Source Code (.py)
4. Streamlit Application
5. Project Report (15–20 pages)
6. Presentation (10–12 slides)
7. GitHub Repository

15. Suggested Report Format

Chapter 1

Introduction

Chapter 2

Literature Review

Chapter 3

Problem Statement

Chapter 4

Dataset Description

Chapter 5

Methodology

Chapter 6

Data Preprocessing

Chapter 7

Exploratory Data Analysis

Chapter 8

Machine Learning Models

Chapter 9

Model Evaluation

Chapter 10

Results and Discussion

Chapter 11

Conclusion

Chapter 12

Future Scope

16. Future Enhancements

Students may extend the project by:

- Predicting startup valuation.
- Recommending suitable investors.
- Predicting funding requirements.
- Performing sentiment analysis on startup news.
- Building an AI-powered investment recommendation dashboard.
- Integrating real-time startup data through APIs.

17. Evaluation Rubric (100 Marks)

Component	Marks
Dataset Collection & Quality	15
Data Preprocessing	15
Exploratory Data Analysis	15
Machine Learning Model Implementation	20
Model Evaluation & Comparison	15
Streamlit Deployment	10
Project Report	5
Presentation & Viva	5

18. Expected Outcome

At the end of the project, students will have developed a complete Machine Learning application capable of predicting the likelihood of a startup surviving for at least five years. The project should demonstrate the entire ML pipeline—from data collection and preprocessing to model training, evaluation, deployment, and interpretation—while also providing practical business insights for investors and entrepreneurs.